

DSCI 510: Final Project Report:

Title: Book Price Analysis Across Platforms

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2. Description:

Online book prices can vary a lot across different websites, but those differences aren't always obvious when someone is deciding where to buy. In this project, we conduct a book price analysis by collecting listings for the same books across multiple sources and matching them using ISBN-13 identifiers. We scrape and retrieve raw price data, then clean and standardize it into a consistent dataset that supports fair comparisons across platforms. Using this processed dataset, we compute summary statistics to compare pricing levels and variability, and identify which website tends to offer lower prices for a given ISBN. Finally, we present our findings using a small set of visualizations that highlight price distributions and cross-site differences.

3. Data:

3.1) Sources of Data:

We collected data from three publicly available web resources to support a cross-site comparison of book prices for the same titles. The overall idea was to (1) generate a large, diverse ISBN seed list using an API source, and (2) use those ISBNs to scrape pricing listings from online book marketplaces. This design allows us to perform price comparisons by matching books across websites using ISBN-13, which serves as a standardized identifier across sources.

We collected data from the following sources:

1. *Open Library (API) - ISBN Seed List*

We used the Open Library API (subjects/search endpoints) to retrieve book records and extract ISBN-13 identifiers. We implemented subject-based retrieval to gather works across multiple genres (eg: Fantasy, History, Science, etc.), so that the sampled books are not drawn from a single category. These ISBNs were used as the seed set for matching the same books across websites.

- *Open Library (Subjects API):*
`https://openlibrary.org/subjects/{slug}.json?limit={limit}&offset={offset}`
- *Open Library (Search API fallback by subject):*
`https://openlibrary.org/search.json?subject={subject}&limit={limit}&page={page}&fields=isbn,title,author_name,key`

2. *AbeBooks (Web Scraping) - Listings by ISBN*

For each ISBN-13 from the seed list, we queried AbeBooks' ISBN search results page (via the *SearchResults* endpoint) and downloaded the HTML using Python's *'request'*. We implemented robust scraping practices and fixed the delay between requests to reduce the likelihood of rate limiting or request failures. The extracted attributes include: listing title (when available), author (when available), condition, listing price, shipping price (if present), total price (computed), currency (best-effort detection), listing URL, and a UTC scrape timestamp.

- *AbeBooks (ISBN search results page):*
<https://www.abebooks.com/servlet/SearchResults?isbn={isbn13}>

3. *BookFinder (Web Scraping) - Aggregated offers across sellers*

We also scraped BookFinder using ISBN-13 URLs to retrieve price offers for the same books. For each extracted offer, we record: ISBN, book title from the page, the seller/website name, offer type derived from condition: new vs used, edition, condition text, and the extracted price. These raw records are saved to *'data/raw/bookfinder_data.csv'* for cleaning and analysis purposes.

- *BookFinder (ISBN page with USD and destination filters):*
<https://www.bookfinder.com/isbn/{isbn13}/?binding=ANY&condition=ANY¤cy=USD&destination=US>

3.2) Number of Data Samples Collected:

We define 'data sample' at each stage as follows:

- Seed samples (*Open Library*): 1,000 ISBN-13 values collected (1 sample = 1 ISBN-13).
- Marketplace listing samples (*AbeBooks*): 726 listings saved (1 sample = 1 scraped listing/offer).
- Marketplace listing samples (*BookFinder*): 1000 listings (1 sample = 1 scraped listing/offer).

Note on collection constraints: While collecting data, we observed some rate limiting on marketplace endpoints (e.g., HTTP 429 responses on AbeBooks), which reduced coverage for certain ISBNs. We addressed this by using polite delays, retries, and continuing with successfully collected listings while documenting the limitation as part of our data collection process.

4. Data Cleaning, Analysis, and Visualization:

4.1) Cleaning the data

To make fair, consistent comparisons across sources, we converted the raw scraped outputs into standardized, analysis-ready tables. The raw files contain multiple offers per ISBN, inconsistent fields across websites, missing values, and occasional parsing artifacts. Our cleaning process focused on (1) selecting comparable variables, (2) validating and converting data types, (3) removing invalid or duplicate records, and (4) producing one representative row per ISBN for each source so that ISBN-level comparisons are meaningful.

We began by loading the AbeBooks raw listings dataset (*data/raw/abebooks_raw.csv*), which contains one row per scraped offer. Since our analysis centers on pricing comparisons, we retained only the columns required for analysis: source, ISBN, author, total_price, and currency. This step reduces noise from inconsistently available fields (e.g., partial title strings or missing shipping values in some listings) and ensures the cleaned table focuses on comparable pricing information.

Next, we addressed the presence of multiple listings for the same ISBN. We sorted listings by total price and then removed duplicates by ISBN, keeping a single representative record per ISBN according to our selection rule. The cleaned dataset was then reorganized to place the ISBN first for readability and saved to *data/processed/abebooks_cleaned.csv*.

BookFinder aggregates offers from many sellers and websites on a single ISBN page, which means the raw output naturally contains multiple seller entries per ISBN. To create a consistent comparison subset, we filtered the BookFinder raw dataset (*data/raw/bookfinder_data.csv*) to retain only offers where the extracted seller name (*raw_website_name*) matched Amazon.com (case-insensitive). This produced a seller-specific dataset that is easier to compare to a marketplace source like AbeBooks.

After filtering, we selected the relevant fields (isbn, raw_website_name, book_title, price) and renamed columns to a common schema (isbn, source, book_title, price). We then removed rows with invalid prices. This included missing values, non-numeric strings, non-positive values, and cases where parsing errors produced 13-digit numbers that resembled ISBNs rather than actual prices. Once invalid entries were removed, we converted the price column into a numeric type to ensure valid comparisons and summary statistics.

Finally, because the same ISBN could still appear more than once (e.g., duplicate entries or repeated offer blocks), we deduplicated the dataset by ISBN. We applied a consistent sorting-and-deduplication rule and retained one representative offer per ISBN. The final cleaned BookFinder subset was saved to *data/processed/amazon_cleaned.csv*.

4.3) Premise

Problem Statement: How do book prices differ across major e-commerce platforms, and which platform consistently offers the best value to consumers?

4.4) Analysis:

After completing data collection and cleaning, we conducted a series of structured analyses to understand how book prices vary between AbeBooks and Amazon, which platform tends to offer better value, and whether pricing patterns differ between new and used books. Our analysis is divided into three major components (Q1–Q3). Below, we describe each step of the code and the logic behind it.

- Loading and Preparing the Data
- Q1 - How Do Prices Differ Across Sources?

We grouped the data by source (AbeBooks vs. various Amazon storefronts) and calculated count, mean & median prices, standard deviation (price volatility), minimum & maximum, 25th and 75th percentiles, Interquartile Range (IQR), Coefficient of Variation (CV) - a normalized measure of price spread

This step allowed us to quantify overall pricing levels and variability across platforms. To evaluate whether the observed price differences were meaningful, we ran an independent t-test comparing AbeBooks prices with Amazon prices. A p-value < 0.05 indicated that the price gap is statistically significant rather than random. We further segmented prices by both source and offer_type, which revealed how pricing differs not only between platforms but also between new and used listings.

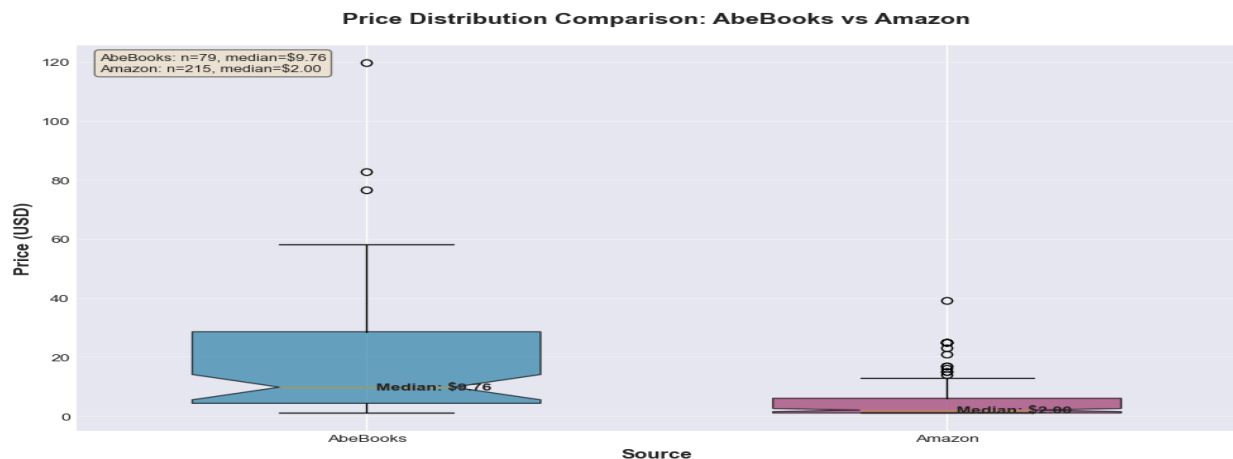
- Q2 - Which Source Offers the Best Value Most Consistently?

We start by identifying books sold on both platforms. To ensure a fair comparison, we restricted analysis to ISBNs that appeared in both datasets. We then filtered the dataset to only these shared ISBNs. For each shared ISBN, we calculated the lowest available price from each platform. This ensures we are comparing the best available deal a customer could reasonably select. We computed the price difference (AbeBooks - Amazon) and labeled the cheaper source. This allowed us to quantify:

- how often is each platform cheaper
 - average savings when a platform "wins"
 - the magnitude of the largest price gaps
- Q3 - Do Pricing Patterns Differ for New vs. Used Books?

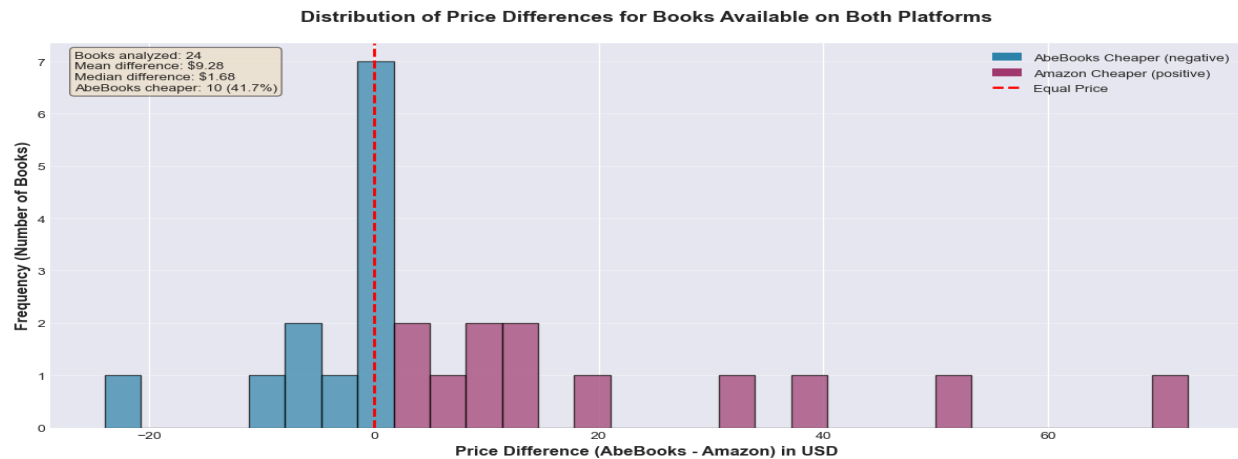
Here, we examined the average price, median price, volatility, and the coefficient of variation

4.5) Visualizations and Interpretations:



Visualization 1 – Price Distribution by Source

Figure X shows a side-by-side boxplot comparing the distribution of total book prices on AbeBooks and Amazon. Each box summarizes the spread of prices for all listings from that source (AbeBooks: n = 79, Amazon: n = 215), with the median price annotated directly on the plot (AbeBooks: \$9.76; Amazon: \$2.00). The AbeBooks box is taller and has several high outliers, indicating both higher typical prices and much greater variability. In contrast, the Amazon distribution is tightly concentrated near the bottom of the range, showing that most Amazon listings in our sample are clustered at low prices.



Visualization 2 – Histogram of Price Differences (AbeBooks – Amazon)

Figure Y plots a histogram of per-book price differences for the 24 ISBNs that appear on both platforms. For each shared ISBN, we compute the minimum available price on AbeBooks and Amazon and then take the difference (AbeBooks - Amazon). Bars to the left of zero (colored for AbeBooks) represent books where AbeBooks is cheaper; bars to the right (colored for Amazon) correspond to cases where Amazon is cheaper, and the dashed red line at zero marks equal pricing. The statistics box on the plot reports that 10 out of 24 books (41.7%) are cheaper on AbeBooks, while the mean difference is \$9.28 in favor of Amazon overall, illustrating that although both platforms sometimes offer the better deal, Amazon's price advantage is typically larger in magnitude.

4.6) Conclusions

Initial Premise: We started with the assumption that book prices vary meaningfully across online platforms, and that these differences could be systematically measured using ISBN-matched listings.

Hypotheses:

1. Amazon will generally offer lower prices than AbeBooks.
2. Used books will be significantly cheaper than new books.
3. Price variability will differ across platforms due to their differing seller ecosystems.

Our findings support all three hypotheses:

1. Amazon is the cheaper platform both on average and for most shared ISBNs.
2. Price differences between platforms are statistically significant.
3. New vs. used pricing differences are large and significant, aligning with typical market expectations.

AbeBooks displays much higher price variability, consistent with its multi-seller marketplace model.

Overall, the analysis demonstrates clear, data-backed differences in book pricing across sources and provides actionable insights into where consumers are more likely to find lower prices. These findings demonstrate that platform choice has a substantial impact on what consumers pay for the same book, and the market behaves differently across sellers. Amazon dominates lower-priced segments, while AbeBooks

shows wider variability and occasional high-value outliers. These conclusions validate the project's premise and illustrate the usefulness of ISBN-based price comparison pipelines for understanding online marketplace dynamics.

5. Challenges Faced and Changes from the Original Proposal:

In our original proposal, we planned a broad e-commerce price comparison by scraping product listings from major retailers such as Walmart, Target, and eBay, and then analyzing prices, discounts, and ratings across platforms. During implementation, we made several adjustments to keep the project feasible and to ensure the final dataset supported reliable comparisons.

First, we narrowed the scope to books as comparing 'general products' across large retailers was too broad for the project timeline and introduced major matching issues (the same product can appear under different names, bundles, or sellers). By focusing on books, we were able to define a clearer unit of comparison and build a more consistent pipeline.

We also changed data sources due to scraping feasibility. Scraping Walmart/Target/eBay reliably can be challenging because these sites often rely on dynamically generated pages and anti-scraping measures, which can result in incomplete HTML, inconsistent parsing, or blocked requests. To avoid paid scraping services, we switched to a workflow that combines Open Library for ISBN seed generation and ISBN-based matching with marketplace sites such as AbeBooks and ThriftBooks, which better supported static scraping for our use case. These changes helped us preserve the original goal of cross-platform price comparison while making the data collection and matching process more stable.

6. Future Work:

With more time and resources, we would extend this project in several directions. First, we would scale up the dataset by collecting more ISBNs across a wider range of genres so the results are less sensitive to what happens to be in our initial sample. We would also add additional retailers and marketplaces to broaden the comparison and capture more variation in pricing behavior. Second, we would improve the real-world accuracy of comparisons by tracking total cost (price, shipping and taxes when available) rather than listing price alone, because that is closer to the price the consumer usually pays. Third, we would run the scraper on a schedule to create a time-series price dataset, enabling analysis of price volatility, discount frequency, and seasonal patterns. To make the results more usable, we would build an interactive dashboard (e.g., Streamlit) where users can search by ISBN or title and view the cheapest source and the full price distribution. Finally, after collecting enough historical data, we could explore machine learning models to predict when discounts are more likely to appear on each platform based on past pricing behavior and timing trends.